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Transmission breather system

Abstract

A motorcycle with a single transmission breather that is mounted over the transmission case. Crank case gas passes through the breather. Oil separated from the breather drains back to a cavity under the breather. Gases separated in the breather travel through a breather hose to an airbox. The breather utilizes a sloped incline in order to prevent oil that separates out from pooling.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit to U.S. provisional application 63/498,196 filed Apr. 25, 2023, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) One or more embodiments relate generally to a transmission breather system for a motorcycle that uses an internal combustion engine.

BACKGROUND

(2) In prior art motorcycles, transmission breathers are used to vent crankcase pressures. These crankcase gases contain oil mist, water vapor, and unburned fuel. The breather system separates the liquids so that the crankcase gases can be introduced into the combustion system. Two breathers can be used for the same motorcycle, one mounted above the heads (front and rear) under rocker covers.

BRIEF SUMMARY

(3) In accordance with one or more embodiments, an example motorcycle is provided that comprises a transmission case; and a breather assembly mounted on the transmission case.

(4) In accordance with one or more embodiments, an example motorcycle is provided that comprises a transmission case; and a breather assembly mounted over the transmission case.

(5) In accordance with the example motorcycle, the breather assembly is the only breather on the motorcycle.

(6) In accordance with the example motorcycle, a plurality of fasteners fastening the breather assembly to the transmission case.

(7) In accordance with the example motorcycle, the breather assembly further comprises: a drain; and a sloped floor connected to the drain.

(8) In accordance with the example motorcycle, the breather assembly further comprises a base,

(9) In accordance with the example motorcycle, the sloped floor makes an angle with the base of 17 degrees.

(10) In accordance with the example motorcycle, the sloped floor makes an angle with the base in a range of 10 to 30 degrees.

(11) In accordance with the example motorcycle, further comprises a drain; and an umbrella valve attached to the drain.

(12) In accordance with one or more embodiments, an apparatus, comprises: a breather assembly that comprises an outlet on a top of the breather assembly; an inlet on a bottom of the breather assembly; a drain on the bottom of the breather assembly; and a sloped floor leading to the drain.

(13) In accordance with the example apparatus, the breather assembly is configured to separate out oil, water vapor, and fuel from the mixture entering the inlet.

(14) In accordance with the example apparatus, the apparatus further comprises a motorcycle attached to the breather assembly.

(15) In accordance with the example apparatus, the breather assembly is the only breather attached to the motorcycle.

(16) In accordance with the example apparatus, the breather assembly is mounted on a transmission case attached to the motorcycle.

(17) In accordance with the example apparatus, the apparatus further comprises an umbrella valve attached to the drain.

(18) In accordance with the example apparatus, the apparatus further comprises a second passage connecting the inlet to an oil reservoir.

(19) In accordance with the example apparatus, the apparatus further comprises a first passage connecting the inlet to a cam chest.

(20) In accordance with one or more embodiments, a motorcycle comprises a transmission; and a breather assembly mounted over the transmission.

(21) In accordance with the example apparatus, the motorcycle has no other breather other than the breather assembly.

(22) In accordance with the example apparatus, the breather assembly further comprises a sloped

floor.
(23) In accordance with the example apparatus, the breather assembly further comprises a drain connected to the sloped floor.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) The various advantages of the exemplary embodiments will become apparent to one skilled in the art by reading the following specification and appended claims, and by referencing the following drawings, in which:
(2) FIG. **1** illustrates a related art transmission cover;
(3) FIG. **2** illustrates an example inside view showing a labyrinth of a breather assembly, in accordance with one or more embodiments set forth, shown, and described herein;
(4) FIG. **3** illustrates a further example inside view showing a labyrinth of a breather assembly, in accordance with one or more embodiments set forth, shown, and described herein;
(5) FIG. **4** illustrates a top inside view showing a breather assembly, in accordance with one or more embodiments set forth, shown, and described herein;
(6) FIG. **5** illustrates a bottom view showing the breather assembly of FIG. **4**, in accordance with one or more embodiments set forth, shown, and described herein;
(7) FIG. **6** illustrates a side internal view showing the breather assembly of FIG. **4**, in accordance with one or more embodiments set forth, shown, and described herein;
(8) FIG. **7** illustrates a top view showing a cover on the breather assembly of FIG. **4**, in accordance with one or more embodiments set forth, shown, and described herein;
(9) FIG. **8** illustrates a view of the breather assembly and surrounding apparatus, in accordance with one or more embodiments set forth, shown, and described herein;
(10) FIG. **9** illustrates a diagonal view of the breather assembly and the surrounding apparatus, in accordance with one or more embodiments set forth, shown, and described herein;
(11) FIG. **10** illustrates an underside view of the breather assembly and the surrounding apparatus, in accordance with one or more embodiments set forth, shown, and described herein;
(12) FIG. **11** illustrates a bottom of the breather assembly including the inlet and drain with surrounding components transparent, in accordance with one or more embodiments set forth, shown, and described herein;
(13) FIG. **12** illustrates a cross section showing the breather inlet and drain tunnel, in accordance with one or more embodiments set forth, shown, and described herein;
(14) FIG. **13** illustrates a breather assembly and breather hose and surrounding apparatus, in accordance with one or more embodiments set forth, shown, and described herein;
(15) FIGS. **14-16** illustrate a breather assembly installed on a motorcycle, in accordance with one or more embodiments set forth, shown, and described herein;
(16) FIG. **17** illustrates a seal for the breather assembly, in accordance with one or more embodiments set forth, shown, and described herein;
(17) FIGS. **18-19** illustrate an example motorcycle having the breather assembly described herein, in accordance with one or more embodiments set forth, shown and described herein;
(18) FIG. **20** illustrates a baffle that fits in the cavity under the breather inlet, in accordance with one or more embodiments set forth, shown, and described herein;
(19) FIG. **21** illustrates a mounted baffle, in accordance with one or more embodiments set forth, shown and described herein; and
(20) FIG. **22** illustrates an unmounted baffle, in accordance with one or more embodiments set forth, shown, and described herein.
DETAILED DESCRIPTION

(21) In accordance with one or more embodiments set forth, shown, and described herein, a transmission breather assembly is mounted on the transmission case of a motorcycle. A breather for a motorcycle serves to manage the pressure inside a crank case. The breather system will act to separate out liquids such as oil, water, or unburned fuel prior to circulating the breather gases to the airbox. The motorcycle which uses the breather assembly and embodiments described herein is the kind that uses gasoline as fuel.
(22) A single breather can be used instead of multiple breathers. FIG. **1** illustrates a related art transmission cover over a transmission case. This cover is no longer needed, and the breather assembly can be mounted above the transmission. In accordance with one or more embodiments set forth, shown, and described herein, the breather assembly can be mounted above the transmission case. In accordance with one or more embodiments set forth, shown, and described herein, the breather assembly can be mounted over the transmission case in place of a transmission cover. In accordance with one or more embodiments set forth, shown, and described herein, a rear cylinder of the motorcycle will be in front of the breather assembly.
(23) As illustrated in FIG. **2**, an example inside view of a labyrinth of a breather assembly is shown, in accordance with one or more embodiments set forth, shown, and described herein. The breather shown has a vent hose **201** which is not part of the breather circuit. A labyrinth **200** is used to separate out oil.
(24) As illustrated in FIG. **3**, a further example inside view a labyrinth of a breather assembly is shown, in accordance with one or more embodiments set forth, shown, and described herein. The breather assembly shown has a vent hose **301** which is not part of the breather circuit. A labyrinth **300** is used to separate out oil.
(25) As illustrated in FIG. **4**, a top inside view of a breather assembly is shown, in accordance with one or more embodiments set forth, shown, and described herein.
(26) A breather media **402** (such as a filter which can be made out of any suitable material, such as foam, stainless steel mesh, fleece, etc.) and impactor plate **407** (which can be made out of any suitable material, such as metal or plastic and has holes in it) are used to separate oil from the vapors. A breather inlet **401** is where vapors enter the breather assembly. A drain **403** is where the oil separated from the vapor can exit the breather assembly. The drain **403** can have a valve such as an umbrella valve **400**.
(27) As illustrated in FIG. **5**, a bottom view of the breather assembly of FIG. **4** is shown, in accordance with one or more embodiments set forth, shown, and described herein. Shown is the drain **403**, umbrella valve **400**, and inlet **401**. There are four holes **500** adapted to receive a fastener (not shown in FIG. **5**) which is how the breather assembly will be mounted on a transmission case.
(28) As illustrated in FIG. **6**, a side internal view of the breather assembly of FIG. **4** is shown, in accordance with one or more embodiments set forth, shown, and described herein. Shown is the breather media **602**, drain **403**, umbrella valve **400**, outlet **600**. Also shown is the base **603** of the breather assembly. A floor **601** is used to help prevent oil that separated from the vapor from pooling, as the separated oil will fall down the incline. The incline can be a 17 degree slope (relative to the base **603**), although other angles can be used as well (e.g., 10 to 30 degrees, or other range). The oil that separates from the vapor will fall down the sloped floor **601** due to gravity and exit out the drain **403**. This slope helps prevent the oil from pooling. The outlet **600** can lead directly to the outside, or it can be attached to a hose that will lead to the outside, or alternatively the outlet can be connected to hose that can recirculate the air coming out of the outlet.
(29) FIG. **7** illustrates a top view showing a cover on the breather assembly of FIG. **4**, in accordance with one or more embodiments set forth, shown, and described herein. A breather cover **700** covers the top of the breather assembly **701**. The outlet **600** passes through the breather cover **700**. The breather assembly **701** can have any combination of any of the structures described and illustrated herein.
(30) As illustrated in FIG. **8**, a view of the breather assembly and surrounding apparatus is shown,

in accordance with one or more embodiments set forth, shown, and described herein. The motorcycle comprises cam chest area **800**, transmission contents **802** (where the transmission is located), and oil reservoir **804**. As shown in FIG. **8**, the breather assembly is mounted over the transmission contents **802** and over the oil reservoir **804**. Second passage **810** is a communication channel between the breather assembly and the oil reservoir **804** and feeds the breather assembly through the inlet **401**. First passage **811** is a communication channel between the breather assembly and the cam chest area **800** and feeds the breather assembly through the inlet **401**. Drained oil falling out of the drain **403** can pass down through the first passage **811**.

(31) As illustrated in FIG. **9**, a diagonal view of the breather assembly and the surrounding apparatus is shown, in accordance with one or more embodiments set forth, shown, and described herein.

(32) As illustrated in FIG. **10**, an underside view of the breather assembly and the surrounding apparatus is shown, in accordance with one or more embodiments set forth, shown, and described herein.

(33) FIG. **11** illustrates a bottom of the breather assembly including the inlet and drain with surrounding components transparent, in accordance with one or more embodiments set forth, shown, and described herein. Inlet **401** is identified and drain **403** is shown. In accordance with one or more embodiments set forth, shown, and described herein, the drain **403** can be connected to the first passage **811**.

(34) As illustrated in FIG. **12**, a cross section showing the breather inlet and drain tunnel is shown, in accordance with one or more embodiments set forth, shown, and described herein.

(35) As illustrated in FIG. **13**, a breather assembly and breather hose and surrounding apparatus is shown, in accordance with one or more embodiments set forth, shown, and described herein. Shown is breather hose **1300** and airbox **1301**. The breather assembly **701** mounts on the transmission case. The crank case gas passes through the breather assembly. The oil separated from the gas in the breather assembly drains back into the cavity under the breather assembly, while the separated crank case gas travels through the breather hose **1300** to the airbox **1301**.

(36) As illustrated in FIGS. **14-16**, a breather assembly installed on a motorcycle is shown, in accordance with one or more embodiments set forth, shown, and described herein. The breather assembly **701** uses four fasteners **1400** to attach the breather assembly **701** to a transmission case **1401** or other structure the breather assembly **701** can be mounted on.

(37) As illustrated in FIG. **17**, a seal for the breather assembly is shown, in accordance with one or more embodiments set forth, shown, and described herein. The seal **1700** can be a PIP (press in place) seal and is used to seal the breather assembly to the transmission case **1401**. The transmission case **1401** (or other mounting structure) has four holes that the fasteners **1400** utilize to secure the breather assembly **701** to the transmission case **1401**.

(38) As illustrated in FIGS. **18-19**, the breather assembly installed on a motorcycle is shown, in accordance with one or more embodiments set forth, shown and described herein. Note that the breather assembly **701** is over the transmission **1801**.

(39) FIG. **20** illustrates a baffle that fits in the cavity under the breather inlet, in accordance with one or more embodiments set forth, shown, and described herein.

(40) The baffle **2000** is shown inserted under the breather **701**. During hard brake or acceleration events, the baffle slows (damps) the motion of the oil entering the cavity under the breather through passages **810** and **811** from the oil reservoir **804** in order to prevent it from entering the breather inlet. The oil would slow by coming into contact with a set of alternating passages which create a labyrinth or a torturous path, thus requiring the oil to flow back and forth a number of times until it reaches the breather inlet **401**.

(41) FIG. **21** illustrates a mounted baffle in accordance with one or more embodiments set forth, shown and described herein. The baffle **2000** is shown mounted in a cavity which would fit under where the breather would be mounted.

(42) FIG. **22** illustrates an unmounted baffle, in accordance with one or more embodiments set forth, shown, and described herein. The baffle can be made from any hard material, such as hard plastic, etc.
(43) Note that all of the parts described herein can be made from any suitable material for that part, such as plastic (e.g., PVC), hard plastic, metal, aluminum, rubber, stainless steel, etc. All parts can be manufactured using any manufacturing technique, such as injection molding, 3-D printing, casting, machining, etc.
(44) The terms “coupled,” “attached,” or “connected” may be used herein to refer to any type of relationship, direct or indirect, between the components in query. Those skilled in the art will appreciate from the foregoing description that the broad techniques of the exemplary embodiments may be implemented in a variety of forms. Therefore, while the embodiments have been described in connection with particular examples thereof, the true scope of the embodiments should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the drawings, specification, and following claims.

Claims

1. An apparatus, comprising: a breather assembly comprising: an outlet on a top of the breather assembly; an inlet on a bottom of the breather assembly; a drain on the bottom of the breather assembly; a sloped floor leading to the drain; a first passage connecting the inlet to a cam chest; and a second passage connecting the inlet to an oil reservoir.
2. The apparatus of claim 1, wherein the breather assembly is configured to separate out oil from vapor entering the inlet.
3. The apparatus of claim 1, further comprising a motorcycle attached to the breather assembly.
4. The apparatus of claim 3, wherein the breather assembly is the only breather attached to the motorcycle.
5. The apparatus of claim 3, wherein the breather assembly is mounted on a transmission case attached to the motorcycle.
6. The apparatus of claim 1, further comprising a baffle connected to the drain.
7. The apparatus of claim 1, further comprising a baffle removably mounted under the inlet for connection to the drain, the baffle comprising a set of alternating passages that prevent flow of oil into the inlet.
8. A motorcycle, comprising: a cam chest; a transmission case; and a breather assembly mounted on the transmission case, the breather assembly including: an outlet on a top of the breather assembly; an inlet on a bottom of the breather assembly; a drain on the bottom of the breather assembly; a sloped floor leading to the drain; and a first passage connecting the inlet to the cam chest.
9. The motorcycle of claim 8, wherein the breather assembly is the only breather on the motorcycle.
10. The motorcycle of claim 8, further comprising a plurality of fasteners fastening the breather assembly to the transmission case.
11. The motorcycle of claim 8, wherein the breather assembly further comprises a base.
12. The motorcycle of claim 11, wherein the sloped floor makes an angle with the base of 17 degrees.
13. The motorcycle of claim 11, wherein the sloped floor makes an angle with the base in a range of 10 to 30 degrees.
14. The motorcycle of claim 8, further comprising: a baffle connected to the drain, the baffle comprising a set of passages.
15. The motorcycle of claim 8, wherein the breather assembly connects to a breather hose which connects to an airbox.

16. The motorcycle of claim 8, further comprising a baffle removably mounted under the inlet for connection to the drain, the baffle comprising a set of alternating passages that prevent flow of oil into the inlet.
17. A motorcycle, comprising: a cam chest; a transmission; and a breather assembly mounted over the transmission, the breather assembly including: an outlet on a top of the breather assembly; an inlet on a bottom of the breather assembly; a drain on the bottom of the breather assembly; a sloped floor leading to the drain; and a passage connecting the inlet to the cam chest.
18. The motorcycle of claim 17, wherein the motorcycle has no other breather other than the breather assembly.
19. The motorcycle of claim 17, further comprising a baffle removably mounted under the inlet for connection to the drain, the baffle comprising a set of alternating passages that prevent flow of oil into the inlet.